EQUALITY IN THE CRITICAL EXPONENT INEQUALITY FOR THE RIESZ PROJECTION

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ABSTRACT. Fu and Li recently proved that the Riesz projection is contractive from L^q to L^p, $p = 4(1 - 1/q)$, for $1 < q < \infty$. We show that equality holds for a nontrivial ψ only if ψ is a constant multiple of an inner function. This confirms a conjecture of Brevig, Llinares and Seip for $1 < q < \infty$, $q \neq 2$; with the known case $q = \infty$, the conjecture is settled.

1. INTRODUCTION

Let m be normalized Lebesgue measure on $\mathbb{T}$, and let P_+ be the Riesz projection, which keeps the Fourier coefficients of nonnegative index. For $1 < q \leq \infty$ let $p(q)$ be the supremum of the p such that $\|P_+\psi\|_p \leq \|\psi\|_q$ for every $\psi \in L^q$. Marzo and Seip [5] showed that $p(\infty) = 4$, and Brevig, Ortega-Cerdà, Seip and Zhao proved $p(q) \leq 4(1 - 1/q)$ [3, Theorem 9] and conjectured equality [3, Conjecture 4]. Brevig, Llinares and Seip [1] settled the endpoint $q = 1$, in the form of an L^0 estimate, and Fu and Li [4] have recently proved the conjecture for every $1 < q < \infty$, so that $p(q) = 4(1-1/q)$:

$$\|P_+\psi\|_p \leq \|\psi\|_q, \qquad p = 4(1 - 1/q). \tag{1}$$

Constant multiples of inner functions give equality in (1). For $q = \infty$ and $q = 4/3$ these are the only extremals [1, Theorem 6 and Corollary 7], while for $q = 1$ there are others [1, Section 3]. Brevig, Llinares and Seip conjectured [1, Section 3] that for $1 < q < 2$ and $2 < q \leq \infty$ equality in $\|P_+\psi\|_{p(q)} \leq \|\psi\|_q$ forces $\psi = CI$ with I inner. For $q = 2$ every $\psi \in H^2$ gives equality.

Theorem 1. *Let $1 < q < \infty$, $q \neq 2$, and $p = 4(1 - 1/q)$. If $\psi \in L^q$ is nontrivial and $\|P_+\psi\|_p = \|\psi\|_q$, then $\psi = CI$ for a constant $C \neq 0$ and an inner function I.*

Together with [1, Theorem 6] for $q = \infty$, this settles the conjecture; for $q = 4/3$ it recovers [1, Corollary 7]. The proof follows that of [4], which bounds a function of $(P_+\psi, \overline{P_-\psi})$ from below by a subharmonic function of

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their product. We observe that this minorant is strictly subharmonic away from the origin, which forces the product to vanish in the case of equality.

2. The minorant of Fu and Li

Throughout, $1 < q < \infty$, $q \neq 2$, $p = 4(1 - 1/q)$, $Q = q/2$, $P = p/2$ and $\lambda = p/q = P/Q$. Then $1 - \lambda = (q-2)^2/q^2$, so $0 < \lambda < 1$, and $PQ = 2Q - 1$, $Q - P = (Q-1)^2/Q > 0$. Put

$$F(x,\zeta) = \lambda \frac{|x+\zeta|^{2Q}}{x^Q} + 1 - \lambda - x^P, \qquad G(\zeta) = \inf_{x>0} F(x,\zeta),$$

for $x > 0$ and $\zeta \in \mathbb{C}$, and $\Phi(w,z) = \lambda|w+z|^q + 1 - \lambda - |w|^p$ for $w, z \in \mathbb{C}$.

Lemma 1.

(1) *$G(0) = 0$, $G(\overline{\zeta}) = G(\zeta)$, and $\Phi(w,z) \geq G(z\overline{w})$ for all $w, z \in \mathbb{C}$.*
(2) *G is smooth on $\mathbb{C}$, and $\partial\overline{\partial}G(\zeta) > 0$ for $\zeta \neq 0$.*
(3) *There is C such that $|G(\zeta)| \leq C(1 + |\zeta|)^Q$ for all $\zeta \in \mathbb{C}$.*

Proof. (1) By the weighted inequality of arithmetic and geometric means, $\lambda x^Q + 1 - \lambda \geq x^{\lambda Q} = x^P$, with equality at $x = 1$; hence $G(0) = 0$. The symmetry holds since $|x + \overline{\zeta}| = |x + \zeta|$ for real x. If $w \neq 0$, put $x = |w|^2$ and $\zeta = z\overline{w}$. Then $|w+z|^q = |x+\zeta|^{2Q}x^{-Q}$ and $|w|^p = x^P$, so $\Phi(w,z) = F(x,\zeta) \geq G(\zeta)$. If $w = 0$, then $\Phi(0,z) \geq 0 = G(0)$.

(2) By [4, Lemma 3.1] there is a unique minimizer $x_{\min}(\zeta) > |\zeta|$, it depends smoothly on ζ, and $\partial_x^2 F > 0$ at $(x_{\min}(\zeta), \zeta)$. Hence $G(\zeta) = F(x_{\min}(\zeta), \zeta)$ is smooth. The computation in [4, Section 3] gives, with $x = x_{\min}(\zeta)$,

$$\partial\overline{\partial}G(\zeta) = \frac{P^3 Q\, x^{2P-2}|\zeta|^2}{(x^2 - |\zeta|^2)^2\, \partial_x^2 F(x,\zeta)},$$

which is strictly positive for $\zeta \neq 0$. This is the last display of [4, Section 3], where the numerator is written $P^2 x^{2P-2} \cdot PQ|\zeta|^2$.

(3) The minimizer satisfies $R(x_{\min}) = 1$, where $R(x) = |x+\zeta|^{2Q-2}(x^2 - |\zeta|^2)x^{-P-Q}$ [4, (11)]. For $x \geq 2|\zeta|$ we have $x/2 \leq |x+\zeta| \leq 3x/2$ and $x^2 - |\zeta|^2 \geq 3x^2/4$, so $R(x) \geq c\, x^{Q-P}$ with $c = (3/4)\min\{2^{2-2Q}, (3/2)^{2Q-2}\}$. Hence either $x_{\min} < 2|\zeta|$, or $1 \geq c\, x_{\min}^{Q-P}$. Since $Q > P$, in both cases $x_{\min}(\zeta) \leq K(1 + |\zeta|)$ with $K = \max\{2, c^{-1/(Q-P)}\}$. Therefore

$$-K^P(1+|\zeta|)^P \leq 1 - \lambda - x_{\min}(\zeta)^P \leq G(\zeta) \leq F(1+|\zeta|, \zeta) \leq \lambda 4^Q (1+|\zeta|)^Q + 1,$$

and $P < Q$. □

3. Proof of Theorem 1

We may assume $\|\psi\|_q = 1$, so that $\|P_+\psi\|_p = 1$. Since $1 < q < \infty$, the theorem of M. Riesz lets us write $\psi = g + \overline{h}$ with $g = P_+\psi \in H^q$ and $h \in H^q$, $h(0) = 0$. Since $p < q$, the function $\Phi(g, \overline{h})$ is integrable on $\mathbb{T}$, and

$$\int_{\mathbb{T}} \Phi(g, \overline{h})\, \mathrm{d}m = \lambda\|\psi\|_q^q + 1 - \lambda - \|g\|_p^p = 0.$$

Let $u = gh$, which lies in H^Q and satisfies $u(0) = 0$. By Lemma 1 (1), $\Phi(g, \overline{h}) \geq G(\overline{h}\overline{g}) = G(u)$ almost everywhere on $\mathbb{T}$, so $\int_{\mathbb{T}} G(u)\,\mathrm{d}m \leq 0$.

The function $v = G \circ u$ is smooth on $\mathbb{D}$, and $\Delta v = 4(\partial\overline{\partial}G)(u)|u'|^2 \geq 0$ by Lemma 1 (2). Let $M(r) = \int_{\mathbb{T}} v(r\xi)\,\mathrm{d}m(\xi)$ for $0 \leq r < 1$. By Green's formula, with A Lebesgue area measure,

$$M(r) - v(0) = \frac{1}{2\pi} \int_{|z|<r} \Delta v(z) \log \frac{r}{|z|}\,\mathrm{d}A(z),$$

so M is nondecreasing and $M(0) = G(0) = 0$. By Lemma 1 (3), $|v(r\xi)| \leq C(1 + u^*(\xi))^Q$, where u^* is the radial maximal function of u, which lies in L^Q since $u \in H^Q$ [2, Chapter 1]. Since u has radial limits almost everywhere and G is continuous, dominated convergence gives $M(r) \to \int_{\mathbb{T}} G(u)\,\mathrm{d}m \leq 0$ as $r \to 1$. Hence $M \equiv 0$, and so $\Delta v \equiv 0$ on $\mathbb{D}$.

Suppose that $u \not\equiv 0$. Since $u(0) = 0$, u is not constant, so the zeros of u and of u' are isolated, and there is $z \in \mathbb{D}$ with $u(z) \neq 0$ and $u'(z) \neq 0$. There $\Delta v(z) > 0$ by Lemma 1 (2), a contradiction. Hence $gh \equiv 0$. Since $\|g\|_p = 1$, $g \not\equiv 0$, and therefore $h \equiv 0$ and $\psi = g$.

Now $\Phi(g, 0) = \lambda|g|^q + 1 - \lambda - |g|^{\lambda q}$ is nonnegative by the weighted inequality of arithmetic and geometric means and has integral 0, so it vanishes almost everywhere. Since $0 < \lambda < 1$, equality in that inequality forces $|g| = 1$. So g is an inner function, and undoing the normalization gives $\psi = CI$. □

4. Concluding remark

Besides the growth bound, the argument only uses that the minorant is strictly subharmonic off the origin. For $q = 2$ the infimum defining G is not attained when $\zeta \neq 0$, and $G(\zeta) = 2\,\mathrm{Re}\,\zeta$ is harmonic; correspondingly every $\psi \in H^2$ is extremal.

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